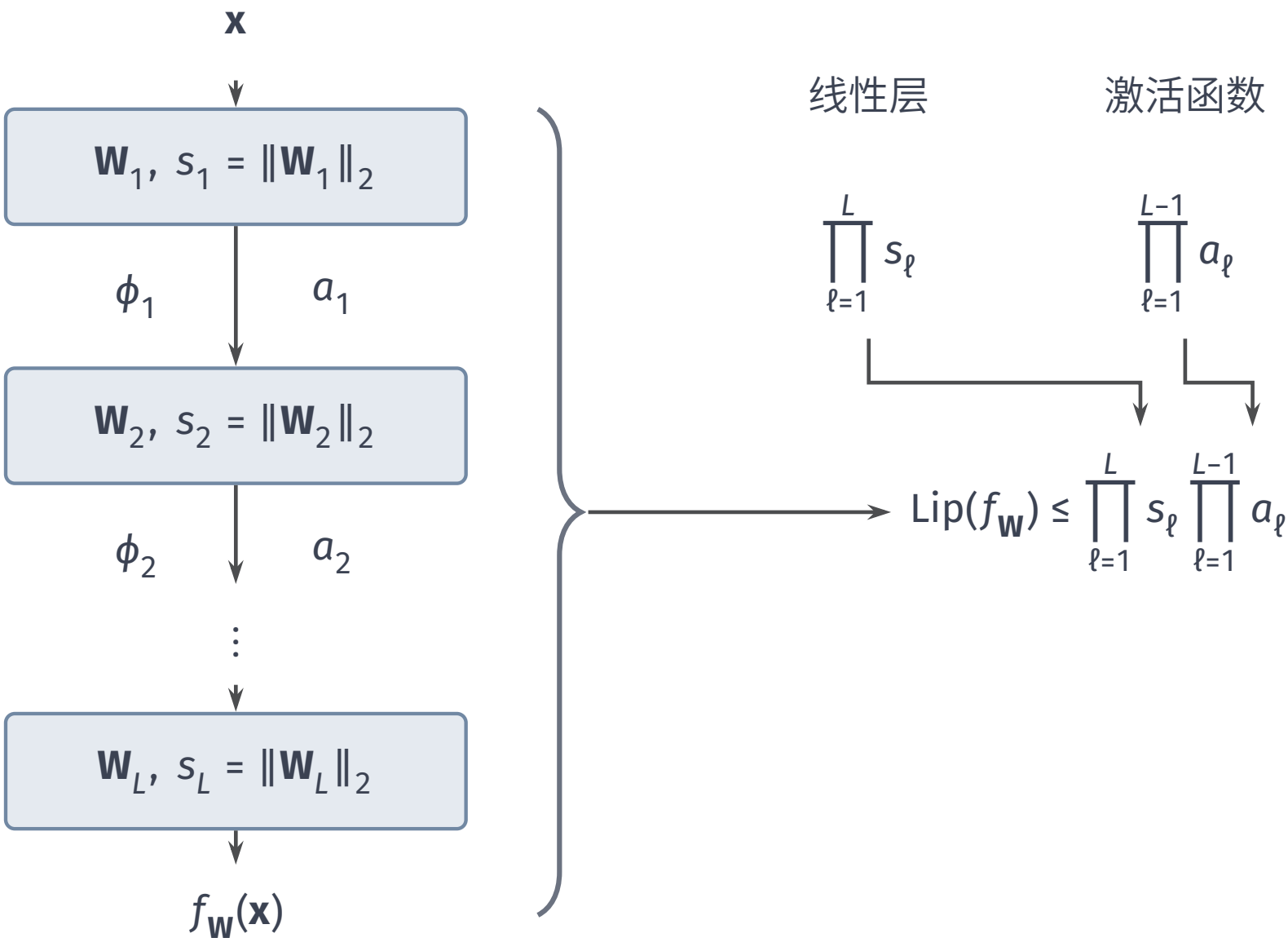

x
$\mathbf{W}_1,\ s_1 = \|\mathbf{W}_1\|_2$
ϕ_1
a_1
$\mathbf{W}_2,\ s_2 = \|\mathbf{W}_2\|_2$
ϕ_2
a_2
$\mathbf{W}_L,\ s_L = \|\mathbf{W}_L\|_2$
$f_{\mathbf{W}}(\mathbf{x})$
线性层
激活函数
$\prod_{\ell=1}^{L} s_\ell$
$\prod_{\ell=1}^{L-1} a_\ell$
$\mathrm{Lip}(f_{\mathbf{W}}) \le \prod_{\ell=1}^{L} s_\ell \prod_{\ell=1}^{L-1} a_\ell$